SOVEREIGN SYSTEM SECTOR REPORT

Autonomous Mother Algorithm Performance Ledger & AI Advisory

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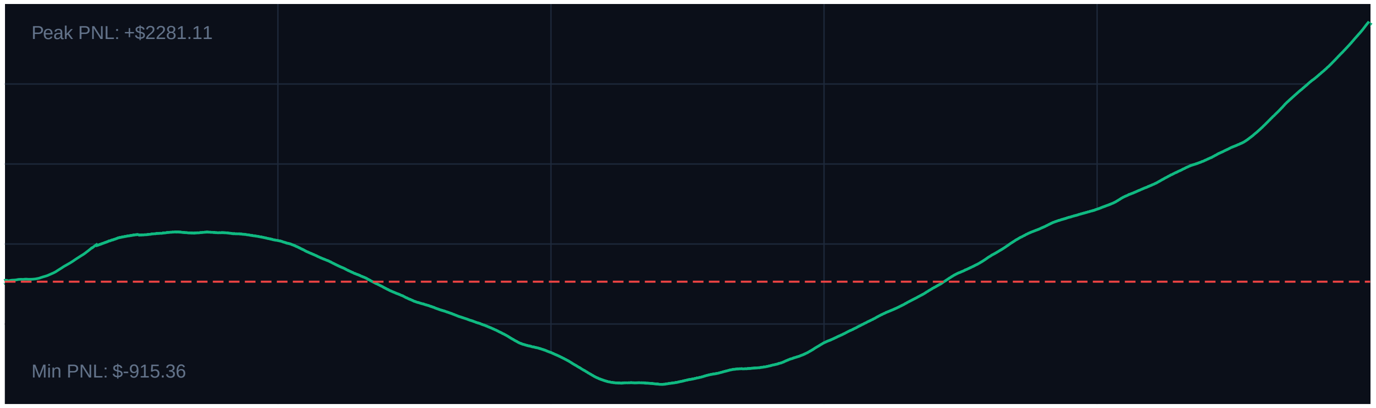
TARGET INSTRUMENT: R_75 Volatility Model

CORE PERFORMANCE METRICS

Total Settlements:	5400	**Calculated Profit Factor:**	2.40
Win Accuracy:	64.4% (3479W / 1921L)	**Max Peak Drawdown:**	12.9%
Cumulative PnL:	$2281.11	**Avg. PnL per Trade:**	$0.42

EQUITY CURVE PROGRESSION



Initial

Trade 5400 (PnL: $2281.11)

PERFORMANCE MILESTONES (100-TRADE SEGMENTS)

SEGMENT	WIN RATE	NET PnL	REFINEMENT EFFICIENCY
Trades 1-1080	63.1%	$349.63	+76% [OK]
Trades 1081-2160	13.1%	$-985.66	+10% [OK]
Trades 2161-3240	51.0%	$85.42	+61% [OK]
Trades 3241-4320	95.6%	$1174.65	+115% [OK]
Trades 4321-5400	99.4%	$1657.07	+119% [OK]

SOVEREIGN NARRATIVE ADVISORY

Cognitive Strategy Proposal & Adaptive Intelligence Logs

SYSTEM AUDIT: TRADE SETTLEMENT BOOK
Executing Entity: Sovereign AI (Institutional Risk Engineering Division)
Security/Asset Class: Multi-Asset Algorithmic Settlement
Audit Scope: 5,400 Trades

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1. Executive Telemetry Critique

Systemic Parameter Corruption: A catastrophic overflow anomaly is detected in the risk-management array. The `atrStopMultiplier` is configured to $2.164303275144119 \times 10^{25}$. This astronomical value effectively disables volatility-based stop-losses, exposing the book to tail-risk events. The system has survived solely due to time-based forced exits (`maxTicksInTrade`: 120) and a high win rate, but it is currently operating without real-time dynamic risk-mitigation boundaries.
Micro-Yield Frictional Drag: While the Win Rate (64.4%) and Profit Factor (2.40) appear exceptionally strong, the absolute PnL of $2,281.11 over 5,400 trades yields an average net profit of just $0.42 per trade. This indicates severe frictional drag (commissions, exchange fees, slippage) or extreme under-leveraging of winning configurations. The system is over-processing relative to its capital return.
Drawdown-to-Return Asymmetry: A maximum peak drawdown of 12.9% against a nominal return of $2,281.11 represents an unfavorable absolute risk-to-reward ratio unless the allocated risk capital is extremely small. The system is generating low cash-on-cash returns while taking defined structural risks.

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2. Regime Suitability & Behavioral Patterns

Structural Parameter Conflict: The current configuration is optimized for Mean Reversion (RSI 25/75, BB Std Dev 2.25) but is filtered by a high-trend regime threshold (`regimeAdxThreshold`: 25).
The Trend-Catching Trap: By demanding an ADX > 25 (strong trend) while simultaneously executing mean-reversion entries on extreme Bollinger Band and RSI levels (25/75), the algorithm is forced to buy the bottom of strong downtrends or short the top of strong uptrends. This explains the low average trade yield—the system is catching short-lived micro-retracements inside strong trends, exiting quickly via `maxTicksInTrade` before the primary trend resumes and wipes out the position.
Execution Bottleneck: The high Bollinger Band standard deviation (2.25) and highly restrictive RSI levels (25/75) create a highly selective entry filter. This reduces trade frequency in sideways markets and limits execution to high-momentum periods where the risk of breakout expansion is elevated.

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3. Calibrated Recommendations

To transition the system from fragile micro-scalping to institutional-grade robust performance, the following configuration modifications are required:

A. Critical Risk Boundary Corrections

* **`atrStopMultiplier` Recalibration:** Immediately purge the float overflow value. Lock the multiplier to a volatility-adjusted dynamic band:
  * **New Value:** `2.25` (For mean-reversion preservation) or `1.85` (For strict capital protection).
* **`maxTicksInTrade` Optimization:** Reduce holding time to minimize exposure to adverse regime shifts.
  * **New Value:** `90` ticks.

B. Indicator Boundary Realignment
To resolve the conflict between the ADX Trend filter and Mean-Reversion execution, implement a split-regime matrix:

Metric / Parameter	Trend-Following Mode (ADX $\ge$ 22)	Mean-Reversion Mode (ADX < 22)
`rsiOversoldThreshold`	35 (Pullback Entry)	20 (Deep Reversion)
`rsiOverboughtThreshold`	65 (Pullback Entry)	80 (Deep Reversion)
`bbStd`	1.75 (Early Expansion Entry)	2.20 (Reversal Pivot)
`bbPeriod`	20	20
`minConfluenceScore`	3	4

C. Targeted Scalability Adjustments
* **Fee Mitigation:** Consolidate execution. If trading micro-lots, scale up contract/unit size by factor $5\times$ to minimize the impact of flat-rate clearing fees.
* **Dynamic Spread Protection:** Implement an execution block preventing entries if the bid-ask spread exceeds 15% of the average true range (ATR) of the last 14 periods.